

How to use this sheet. Copy it by hand in this order. Yellow boxes like this one are notes for you and do not need to be copied. Boxed results are final answers; underline or box them in your handwritten copy too. **Part A** is the general mathematical model for the real dataset, about 4–5 handwritten pages. **Part B** works one forward pass, loss, backward pass and update on a small 5-transaction graph with real numbers, about 6–7 handwritten pages. All numbers were checked with Python and are rounded to 4 decimals.

Vanilla GCN

Mathematical Modelling

Illicit Bitcoin transaction detection on the Elliptic dataset using a 2-layer GCN

[Your name] [roll no.]

PART A: Mathematical Model of the Real Problem

A1. Problem

Binary node classification on a transaction graph: for each Bitcoin transaction v , predict

$$y_v \in \{0, 1\}, \quad 0 = \text{licit}, \quad 1 = \text{illicit},$$

using the transaction's own features *and* the transactions it is connected to.

A2. Graph Representation

$$G = (V, E), \quad |V| = N = 203,769 \text{ transactions}, \quad |E| = 234,355 \text{ money flows}$$

- Edge ($u \rightarrow v$): an output of transaction u is spent as an input of transaction v .
- Adjacency matrix: $A \in \{0, 1\}^{N \times N}$, $A_{uv} = 1$ if u and v are connected.
- Feature matrix: $X \in \mathbb{R}^{N \times F}$, $F = 165$ (93 local + 72 aggregated features).
- Labels: $y_v \in \{0, 1, \text{unknown}\}$; 4,545 illicit, 42,019 licit, 157,205 unknown.
- Time step: $t(v) \in \{1, 2, \dots, 49\}$, about two weeks apart.

Temporal split (train on the past, test on the future):

$$V_{\text{train}} = \{v : y_v \text{ known}, t(v) \leq 34\}, \quad V_{\text{test}} = \{v : y_v \text{ known}, t(v) \geq 35\}$$

$$|V_{\text{train}}| = 29,894 \text{ (3,462 illicit)}, \quad |V_{\text{test}}| = 16,670 \text{ (1,083 illicit)}$$

A3. Making the Graph Undirected

Money flow is directed, but the GCN uses information in both directions:

$$A \leftarrow \max(A, A^T) \quad \Rightarrow \quad A = A^T \text{ (symmetric)}$$

A4. Symmetric Normalisation of the Adjacency Matrix

Self-loops (a node keeps its own features):

$$\tilde{A} = A + I_N$$

Degree matrix:

$$\tilde{D}_{ii} = \tilde{d}_i = \sum_j \tilde{A}_{ij} \quad (\text{number of neighbours} + 1)$$

Symmetric normalisation:

$$\boxed{\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}} \iff \hat{A}_{ij} = \frac{\tilde{A}_{ij}}{\sqrt{\tilde{d}_i \tilde{d}_j}}$$

Why: A alone gives high-degree nodes, such as transactions with hundreds of outputs, much larger sums than low-degree nodes. Dividing by $\sqrt{\tilde{d}_i \tilde{d}_j}$ gives a fair weighted average and keeps $\hat{A}$ symmetric ($\hat{A}_{ij} = \hat{A}_{ji}$).

Sparsity:

$$\text{nnz}(\hat{A}) = 2|E| + N = 2(234,355) + 203,769 = 672,479$$

$$\text{Dense storage} = N^2 \times 4 \text{ bytes} = (203,769)^2 \times 4 \approx 1.66 \times 10^{11} \text{ bytes} \approx 166 \text{ GB}$$

So $\hat{A}$ is stored as a *sparse* matrix with only 672,479 non-zeros.

A5. Feature Standardisation

For every feature j , using only the training period ($t \leq 34$):

$$x_{ij} \leftarrow \frac{x_{ij} - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{|T|} \sum_{i \in T} x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{|T|} \sum_{i \in T} (x_{ij} - \mu_j)^2}, \quad T = \{i : t(i) \leq 34\}$$

A6. GCN Layer (Propagation Rule)

$$\boxed{H^{(l+1)} = \sigma(\hat{A} H^{(l)} W^{(l)} + b^{(l)})}, \quad H^{(0)} = X$$

- $H^{(l)} \in \mathbb{R}^{N \times d_l}$: node features entering layer l
- $W^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$: learnable weight matrix; $b^{(l)}$: bias
- $\hat{A}$: normalised adjacency (fixed, not learned)
- σ : activation (ReLU for the hidden layer, $\text{ReLU}(x) = \max(0, x)$)

For one node v :

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{\tilde{d}_u \tilde{d}_v}} h_u^{(l)} W^{(l)} + b^{(l)}\right)$$

That is, the node averages itself and its neighbours, weighted by degree, then applies the transform.

A7. Two-Layer Forward Pass

$$\boxed{Z = \text{softmax}\left(\hat{A} \text{ReLU}(\hat{A} X W^{(0)} + b^{(0)}) W^{(1)} + b^{(1)}\right)}$$

$N = 203,769$, $F = 165$, $H = 100$ hidden units, $C = 2$ classes.

Step	Shape	Meaning
X	$N \times F$	input features
$\bar{X} = \hat{A}X$	$N \times F$	aggregate neighbours (computed once)
$S = \bar{X}W^{(0)} + b^{(0)}$	$N \times H$	linear transform
$H^{(1)} = \text{ReLU}(S)$	$N \times H$	non-linearity
$H^{(1)}W^{(1)}$	$N \times C$	transform to 2 scores
$Z_{\text{logits}} = \hat{A}(H^{(1)}W^{(1)}) + b^{(1)}$	$N \times C$	aggregate again
$P_{ic} = \frac{e^{Z_{ic}}}{\sum_{c'} e^{Z_{ic'}}}$	$N \times C$	softmax probabilities

Pre-computation trick (by associativity of matrix multiplication):

$$\hat{A}(XW^{(0)}) = (\hat{A}X)W^{(0)}$$

Since $\hat{A}$ and X never change, $\hat{A}X$ is computed once before training.

Number of parameters:

$$\underbrace{165 \times 100 + 100}_{W^{(0)}, b^{(0)}} + \underbrace{100 \times 2 + 2}_{W^{(1)}, b^{(1)}} = 16,500 + 100 + 200 + 2 = \boxed{16,802}$$

A8. Weighted Cross-Entropy Loss

Only about 10% of labelled transactions are illicit, so illicit mistakes are weighted more: $w_0 = 0.3$ (licit), $w_1 = 0.7$ (illicit).

$$\mathcal{L} = -\frac{1}{\sum_{i \in V_{\text{train}}} w_{y_i}} \sum_{i \in V_{\text{train}}} w_{y_i} \log P_{i, y_i}$$

Computed *only* over labelled training nodes. Unlabelled nodes still affect the loss through $\hat{A}$.

Check: a uniform guess $P_{i,c} = \frac{1}{2}$ gives $\mathcal{L} = -\log \frac{1}{2} = \ln 2 \approx 0.6931$.

Effective class weight:

$$0.7 \times 3,462 = 2,423.4 \quad (\text{illicit}) \quad \text{vs} \quad 0.3 \times 26,432 = 7,929.6 \quad (\text{licit})$$

A9. Gradients (Backpropagation)

Let $\Omega = \sum_{i \in V_{\text{train}}} w_{y_i}$ and Y be the one-hot labels.

$$\frac{\partial \mathcal{L}}{\partial Z_i} = \frac{w_{y_i}}{\Omega} (P_i - Y_i) \quad (i \in V_{\text{train}}), \quad 0 \text{ otherwise}$$

$$M = \hat{A}H^{(1)} \Rightarrow Z = MW^{(1)}$$

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} = M^T \frac{\partial \mathcal{L}}{\partial Z}$$

$$\frac{\partial \mathcal{L}}{\partial H^{(1)}} = \hat{A}^T \left(\frac{\partial \mathcal{L}}{\partial Z} W^{(1)T} \right) = \hat{A} \left(\frac{\partial \mathcal{L}}{\partial Z} W^{(1)T} \right) \quad (\hat{A} \text{ symmetric})$$

$$\frac{\partial \mathcal{L}}{\partial S} = \frac{\partial \mathcal{L}}{\partial H^{(1)}} \odot \mathbb{1}[S > 0] \quad (\text{ReLU gate})$$

$$\frac{\partial \mathcal{L}}{\partial W^{(0)}} = (\hat{A}X)^T \frac{\partial \mathcal{L}}{\partial S}$$

A10. Training with Adam

Parameters $\theta = \{W^{(0)}, b^{(0)}, W^{(1)}, b^{(1)}\}$. With gradient $g_t = \nabla_{\theta} \mathcal{L} + \lambda \theta$ (weight decay $\lambda = 5 \times 10^{-4}$):

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \\ \theta_t &= \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \end{aligned}$$

$\eta = 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, 200 epochs, full batch.

One epoch: (1) forward pass on the whole graph $\rightarrow Z$; (2) loss on V_{train} only; (3) backward pass $\rightarrow \nabla_{\theta} \mathcal{L}$; (4) Adam update.

A11. Prediction Rule

$$\hat{y}_i = \begin{cases} 1 \text{ (illicit)} & \text{if } P_{i,1} \geq 0.5 \\ 0 \text{ (licit)} & \text{otherwise} \end{cases}$$

A12. Evaluation Metrics (illicit = positive class)

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN} \\ F_1 &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \\ \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \end{aligned}$$

Our GCN on the test set (seed 42): $TP = 507$, $FP = 182$, $FN = 576$, $TN = 15,405$.

$$\begin{aligned} \text{Precision} &= \frac{507}{507 + 182} = \frac{507}{689} = 0.7358 \\ \text{Recall} &= \frac{507}{507 + 576} = \frac{507}{1083} = 0.4681 \\ F_1 &= \frac{2(507)}{2(507) + 182 + 576} = \frac{1014}{1772} = \boxed{0.5722} \\ \text{Accuracy} &= \frac{507 + 15,405}{16,670} = \frac{15,912}{16,670} = 0.9545 \end{aligned}$$

Why accuracy misleads: always predicting “licit” gives Accuracy = $\frac{15,587}{16,670} = 0.9350$ but $TP = 0 \Rightarrow F_1 = 0$.

A13. Variants Used for Comparison

MLP (no graph): replace $\hat{A}$ by I :

$$Z = \text{softmax}(\text{ReLU}(XW^{(0)})W^{(1)})$$

Skip-GCN: add a direct path from the input:

$$Z = \text{softmax}(\hat{A} \text{ReLU}(\hat{A}XW^{(0)})W^{(1)} + XW_{\text{skip}}), \quad W_{\text{skip}} \in \mathbb{R}^{165 \times 2}$$

A14. Over-smoothing (why we use only 2 layers)

$\hat{A}$ is symmetric, so $\hat{A} = \sum_i \mu_i u_i u_i^T$ with eigenvalues $1 = \mu_1 > |\mu_2| \geq \dots$ on a connected graph. The top eigenvector is

$$\pi = \frac{\tilde{D}^{1/2} \mathbf{1}}{\|\tilde{D}^{1/2} \mathbf{1}\|}, \quad \hat{A} \pi = \pi$$

Applying $\hat{A}$ k times:

$$\hat{A}^k H = \pi \pi^T H + \sum_{i \geq 2} \mu_i^k u_i u_i^T H \xrightarrow{k \rightarrow \infty} \pi \pi^T H$$

since $|\mu_i|^k \rightarrow 0$. Every row becomes a multiple of the same vector $\pi^T H$, so all nodes look alike. Observed on Elliptic (local features): F1 = 0.273, **0.501**, 0.422, 0.385, 0.336 for 1, 2, 3, 4, 6 layers.

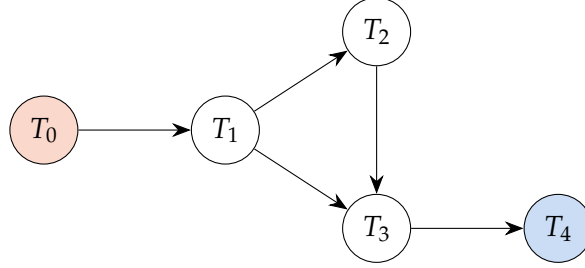
PART B: Worked Example on a Small Transaction Graph

B1. The Graph

Five Bitcoin transactions $V = \{T_0, T_1, T_2, T_3, T_4\}$ with money flows

$$E = \{(0,1), (1,2), (1,3), (2,3), (3,4)\}$$

treated as undirected.



T_0 = known illicit (label 1), T_4 = known licit (label 0), T_1, T_2, T_3 = unknown

Adjacency matrix, degree vector, degree matrix:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad d = [1, 3, 2, 3, 1], \quad D = \text{diag}(1, 3, 2, 3, 1)$$

B2. Add Self-Loops

$$\tilde{A} = A + I = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad \tilde{d} = [2, 4, 3, 4, 2], \quad \tilde{D} = \text{diag}(2, 4, 3, 4, 2)$$

B3. Symmetric Normalisation

$$\tilde{D}^{-1/2} = \text{diag}\left(\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{\sqrt{3}}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right), \quad \hat{A}_{ij} = \frac{\tilde{A}_{ij}}{\sqrt{\tilde{d}_i \tilde{d}_j}}$$

Entries: $\frac{1}{\sqrt{2} \cdot 2} = \frac{1}{2}$, $\frac{1}{\sqrt{2} \cdot 4} = \frac{1}{2\sqrt{2}}$, $\frac{1}{\sqrt{4} \cdot 4} = \frac{1}{4}$, $\frac{1}{\sqrt{4} \cdot 3} = \frac{1}{2\sqrt{3}}$, $\frac{1}{\sqrt{3} \cdot 3} = \frac{1}{3}$.

$$\hat{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2\sqrt{2}} & 0 & 0 & 0 \\ \frac{1}{2\sqrt{2}} & \frac{1}{4} & \frac{1}{2\sqrt{3}} & \frac{1}{4} & 0 \\ 0 & \frac{1}{2\sqrt{3}} & \frac{1}{3} & \frac{1}{2\sqrt{3}} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2\sqrt{3}} & \frac{1}{4} & \frac{1}{2\sqrt{2}} \\ 0 & 0 & 0 & \frac{1}{2\sqrt{2}} & \frac{1}{2} \end{bmatrix} \approx \begin{bmatrix} 0.5 & 0.3536 & 0 & 0 & 0 \\ 0.3536 & 0.25 & 0.2887 & 0.25 & 0 \\ 0 & 0.2887 & 0.3333 & 0.2887 & 0 \\ 0 & 0.25 & 0.2887 & 0.25 & 0.3536 \\ 0 & 0 & 0 & 0.3536 & 0.5 \end{bmatrix}$$

$\hat{A}$ is symmetric, as required, since $\tilde{A}$ is symmetric and $\tilde{D}$ is diagonal.

B4. Feature Matrix

Two features per transaction, $X \in \mathbb{R}^{5 \times 2}$: f_1 = “unusual output pattern” (1 = yes), f_2 = “normal fee pattern” (1 = yes).

$$X = H^{(0)} = \begin{array}{c|cc} \text{Tx} & f_1 & f_2 \\ \hline T_0 & 1 & 0 \\ T_1 & 1 & 1 \\ T_2 & 1 & 0 \\ T_3 & 0 & 1 \\ T_4 & 0 & 1 \end{array} \quad \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

B5. Labels and Loss Weights

$C = 2$ (class 0 = licit, class 1 = illicit). Labelled training nodes $V_L = \{T_0, T_4\}$:

$$y_0 = 1 \Rightarrow Y_0 = [0, 1] \text{ (weight 0.7)}, \quad y_4 = 0 \Rightarrow Y_4 = [1, 0] \text{ (weight 0.3)}$$

T_1, T_2, T_3 are unlabelled during training.

B6. Layer 1: Aggregation $\bar{X} = \hat{A}X$

$$(\hat{A}X)_{ij} = \sum_k \hat{A}_{ik} X_{kj}:$$

$$\bar{X}_0 = \frac{1}{2}(1, 0) + \frac{1}{2\sqrt{2}}(1, 1) = \left(\frac{1}{2} + \frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right) = (0.8536, 0.3536)$$

$$\bar{X}_1 = \frac{1}{2\sqrt{2}}(1, 0) + \frac{1}{4}(1, 1) + \frac{1}{2\sqrt{3}}(1, 0) + \frac{1}{4}(0, 1) = \left(\frac{1}{2\sqrt{2}} + \frac{1}{4} + \frac{1}{2\sqrt{3}}, \frac{1}{2}\right) = (0.8922, 0.5)$$

$$\bar{X}_2 = \frac{1}{2\sqrt{3}}(1, 1) + \frac{1}{3}(1, 0) + \frac{1}{2\sqrt{3}}(0, 1) = \left(\frac{1}{2\sqrt{3}} + \frac{1}{3}, \frac{1}{\sqrt{3}}\right) = (0.6220, 0.5774)$$

$$\bar{X}_3 = \frac{1}{4}(1, 1) + \frac{1}{2\sqrt{3}}(1, 0) + \frac{1}{4}(0, 1) + \frac{1}{2\sqrt{2}}(0, 1) = \left(\frac{1}{4} + \frac{1}{2\sqrt{3}}, \frac{1}{2} + \frac{1}{2\sqrt{2}}\right) = (0.5387, 0.8536)$$

$$\bar{X}_4 = \frac{1}{2\sqrt{2}}(0, 1) + \frac{1}{2}(0, 1) = \left(0, \frac{1}{2} + \frac{1}{2\sqrt{2}}\right) = (0, 0.8536)$$

B7. Layer 1: Transform and ReLU

Weights (bias = 0 for simplicity). Hidden unit 1 = “suspicion” ($f_1 - f_2$), hidden unit 2 = “normality” ($f_2 - f_1$):

$$W^{(0)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad S = \bar{X} W^{(0)} \Rightarrow S_i = (\bar{X}_{i1} - \bar{X}_{i2}, \bar{X}_{i2} - \bar{X}_{i1})$$

$$S = \begin{bmatrix} 0.5 & -0.5 \\ 0.3922 & -0.3922 \\ 0.0447 & -0.0447 \\ -0.3149 & 0.3149 \\ -0.8536 & 0.8536 \end{bmatrix} \xrightarrow{\text{ReLU}} H^{(1)} = \begin{bmatrix} 0.5 & 0 \\ 0.3922 & 0 \\ 0.0447 & 0 \\ 0 & 0.3149 \\ 0 & 0.8536 \end{bmatrix}$$

ReLU sets every negative value to 0: T_0, T_1, T_2 activate “suspicion”; T_3, T_4 activate “normality”.

B8. Layer 2: Aggregation $M = \hat{A}H^{(1)}$

$$M_0 = \frac{1}{2}(0.5, 0) + 0.3536(0.3922, 0) = (0.25 + 0.1387, 0) = (0.3887, 0)$$

$$M_4 = 0.3536(0, 0.3149) + \frac{1}{2}(0, 0.8536) = (0, 0.1113 + 0.4268) = (0, 0.5381)$$

and similarly for the other rows:

$$M = \hat{A}H^{(1)} \approx \begin{bmatrix} 0.3887 & 0 \\ 0.2877 & 0.0787 \\ 0.1281 & 0.0909 \\ 0.1109 & 0.3805 \\ 0 & 0.5381 \end{bmatrix}$$

B9. Layer 2: Output Logits $Z = MW^{(1)}$

Suspicion should push towards illicit and normality towards licit:

$$W^{(1)} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

With column 1 of M = suspicion (s_i) and column 2 = normality (n_i):

$$Z_{i0} \text{ (licit score)} = n_i - s_i, \quad Z_{i1} \text{ (illicit score)} = s_i - n_i = -Z_{i0}$$

$$T_0 : Z_0 = (0 - 0.3887, 0.3887 - 0) = (-0.3887, 0.3887)$$

All rows:

$$Z \approx \begin{bmatrix} -0.3887 & 0.3887 \\ -0.2090 & 0.2090 \\ -0.0372 & 0.0372 \\ 0.2695 & -0.2695 \\ 0.5381 & -0.5381 \end{bmatrix}$$

B10. Softmax

$$P_{ic} = \frac{e^{Z_{ic}}}{e^{Z_{i0}} + e^{Z_{i1}}}$$

Because $Z_{i1} = -Z_{i0}$, the two-class softmax becomes a sigmoid:

$$P_{i1} = \frac{e^z}{e^{-z} + e^z} = \frac{1}{1 + e^{-2z}}, \quad z = Z_{i1}$$

Labelled node T_0 : $z = 0.3887$:

$$P_{0,1} = \frac{1}{1 + e^{-0.7774}} = \frac{1}{1 + 0.4596} = 0.6851, \quad P_{0,0} = 0.3149$$

Labelled node T_4 : $z = -0.5381$:

$$P_{4,1} = \frac{1}{1 + e^{1.0762}} = \frac{1}{1 + 2.9335} = 0.2542, \quad P_{4,0} = 0.7458$$

All nodes:

$$P \approx \begin{bmatrix} 0.3149 & 0.6851 \\ 0.3970 & 0.6030 \\ 0.4814 & 0.5186 \\ 0.6316 & 0.3684 \\ 0.7458 & 0.2542 \end{bmatrix} \quad \begin{array}{l} T_0 : \text{illicit } \checkmark \\ T_1 : \text{illicit (predicted)} \\ T_2 : \text{illicit, borderline} \\ T_3 : \text{licit (predicted)} \\ T_4 : \text{licit } \checkmark \end{array}$$

The unlabelled T_1 is flagged as illicit because it receives money directly from the illicit T_0 .

B11. Weighted Cross-Entropy Loss

$$\mathcal{L} = -\frac{w_1 \log P_{0,1} + w_0 \log P_{4,0}}{w_1 + w_0}$$

$$\ell_0 = -\log(0.6851) = 0.3782, \quad \ell_4 = -\log(0.2542) = 0.2933$$

$$\mathcal{L} = \frac{0.7(0.3782) + 0.3(0.2933)}{0.7 + 0.3} = 0.2647 + 0.0880 = \boxed{0.3527}$$

This is below $\ln 2 = 0.6931$, so the guess is better than random.

B12. Gradient of Softmax Cross-Entropy

$$P_{ic} = \frac{e^{Z_{ic}}}{\sum_{c'} e^{Z_{ic'}}}, \quad \ell_i = -\sum_c Y_{ic} \log P_{ic}$$

$$\frac{\partial P_{ic}}{\partial Z_{ic}} = P_{ic}(1 - P_{ic}), \quad \frac{\partial P_{ic}}{\partial Z_{id}} = -P_{ic}P_{id} \quad (c \neq d)$$

Combining:

$$\frac{\partial \ell_i}{\partial Z_{ic}} = P_{ic} - Y_{ic} \quad \Rightarrow \quad \frac{\partial \mathcal{L}}{\partial Z_i} = \frac{w_{y_i}}{\sum w} (P_i - Y_i)$$

$$T_0 : 0.7 ([0.3149, 0.6851] - [0, 1]) = [0.2204, -0.2204]$$

$$T_4 : 0.3 ([0.7458, 0.2542] - [1, 0]) = [-0.0763, 0.0763]$$

$$dZ = \begin{bmatrix} 0.2204 & -0.2204 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -0.0763 & 0.0763 \end{bmatrix} \quad (\text{unlabelled rows are 0})$$

B13. Backpropagation to $W^{(1)}$

Since $Z = MW^{(1)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} = M^T dZ = M_0^T dZ_0 + M_4^T dZ_4$$

$$= \begin{bmatrix} 0.3887 \\ 0 \end{bmatrix} [0.2204, -0.2204] + \begin{bmatrix} 0 \\ 0.5381 \end{bmatrix} [-0.0763, 0.0763] = \begin{bmatrix} 0.0857 & -0.0857 \\ -0.0410 & 0.0410 \end{bmatrix}$$

B14. Backpropagation to $H^{(1)}$

$$dM = dZ W^{(1)T} : \quad dM_0 = [-0.4408, 0.4408], \quad dM_4 = [0.1525, -0.1525], \quad \text{others } 0$$

Since $M = \hat{A}H^{(1)}$ and $\hat{A}^T = \hat{A}$:

$$dH^{(1)} = \hat{A} dM \Rightarrow dH_i = \hat{A}_{i0} dM_0 + \hat{A}_{i4} dM_4$$

$$dH^{(1)} = \begin{bmatrix} -0.2204 & 0.2204 \\ -0.1559 & 0.1559 \\ 0 & 0 \\ 0.0539 & -0.0539 \\ 0.0763 & -0.0763 \end{bmatrix}$$

T_1 and T_3 get a gradient even though they are unlabelled, because they are neighbours of T_0 and T_4 .

B15. Through ReLU and to $W^{(0)}$

ReLU passes the gradient only where $S > 0$:

$$dS = dH^{(1)} \odot \mathbb{1}[S > 0] = \begin{bmatrix} -0.2204 & 0 \\ -0.1559 & 0 \\ 0 & 0 \\ 0 & -0.0539 \\ 0 & -0.0763 \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial W^{(0)}} = (\hat{A}X)^T dS = \bar{X}^T dS$$

$$[1, 1] : 0.8536(-0.2204) + 0.8922(-0.1559) = -0.3272$$

$$[1, 2] : 0.5387(-0.0539) = -0.0291$$

$$[2, 1] : 0.3536(-0.2204) + 0.5(-0.1559) = -0.1559$$

$$[2, 2] : 0.8536(-0.0539) + 0.8536(-0.0763) = -0.1111$$

$$\frac{\partial \mathcal{L}}{\partial W^{(0)}} = \begin{bmatrix} -0.3272 & -0.0291 \\ -0.1559 & -0.1111 \end{bmatrix}$$

B16. Gradient Descent Update ($\eta = 0.5$)

$$W \leftarrow W - \eta \frac{\partial \mathcal{L}}{\partial W}$$

$$W_{\text{new}}^{(0)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} - 0.5 \begin{bmatrix} -0.3272 & -0.0291 \\ -0.1559 & -0.1111 \end{bmatrix} = \begin{bmatrix} 1.1636 & -0.9855 \\ -0.9221 & 1.0556 \end{bmatrix}$$

$$W_{\text{new}}^{(1)} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - 0.5 \begin{bmatrix} 0.0857 & -0.0857 \\ -0.0410 & 0.0410 \end{bmatrix} = \begin{bmatrix} -1.0428 & 1.0428 \\ 1.0205 & -1.0205 \end{bmatrix}$$

Repeating the forward pass with the new weights:

$$P_{0,1} : 0.6851 \rightarrow 0.7542, \quad P_{4,0} : 0.7458 \rightarrow 0.7661, \quad \mathcal{L} : 0.3527 \rightarrow \boxed{0.2773}$$

The loss decreased, so one full training iteration of the vanilla GCN is complete. The real model repeats this 200 times with Adam on 203,769 transactions.

B17. Over-smoothing on the Small Graph

Apply $\hat{A}$ repeatedly with no weights, $H_k = \hat{A}^k X$, and measure the mean cosine similarity between all pairs of rows:

k	1	2	4	8	16	32
mean cosine similarity	0.7962	0.8922	0.9739	0.9987	1.0000	1.0000

Eigenvalues of $\hat{A}$: 1, 0.6830, 0.4302, -0.0969 , -0.1830 .

Top eigenvector:

$$\pi \propto \sqrt{d} = [\sqrt{2}, 2, \sqrt{3}, 2, \sqrt{2}] \Rightarrow \pi = [0.3651, 0.5164, 0.4472, 0.5164, 0.3651]$$

Limit: $H_\infty = \pi\pi^T X$, with $\pi^T X = [1.3287, 1.3979]$:

$$H_\infty = \begin{bmatrix} 0.485 & 0.510 \\ 0.686 & 0.722 \\ 0.594 & 0.625 \\ 0.686 & 0.722 \\ 0.485 & 0.510 \end{bmatrix}$$

Every row is a multiple of $[1.3287, 1.3979]$. The illicit T_0 and the licit T_4 now have *identical* features, so they cannot be told apart. This is why we use only 2 layers.